

Quantum Entanglement and Quantum Computing

PROBLEM BANK

Mathematical Tools and Dirac Notation

*"I think I can safely say that nobody understands
quantum mechanics."*
— Richard P. Feynman

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TOPICS COVERED:

- Linear Independence in Function and Vector Spaces
- Dirac Notation (Bra-Ket Formalism)
- Matrix Representation of State Vectors
- Hermitian Conjugate Operations
- Hermitian Operators and Their Properties
- Unitary Matrices and Their Characteristics
- Unitary Operators and Transformations
- Physical Interpretation of Scalar Products
- Measurement and Expectation Values

"Mathematics is the language with which God has written the universe."
— Galileo Galilei

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1 LINEAR INDEPENDENCE OF FUNCTIONS

Note: Linear Independence

A set of functions (or vectors) is **linearly independent** if none can be written as a combination of the others. If one can be expressed as a combination of the others, they are **linearly dependent**. This determines whether the set forms a basis for the vector space.

1.1 Problem Set 1.1: Basic Function Independence

Problem 1.1.1

Check whether the following sets of functions are linearly independent or dependent on the real x-axis:

(a) $f(x) = 4, g(x) = x^2, h(x) = e^{2x}$

(b) $f(x) = x, g(x) = x^2, h(x) = x^3$

(c) $f(x) = x, g(x) = 5x, h(x) = x^2$

Solution 1.1.1

Method: Functions are linearly independent if $a_1f(x) + a_2g(x) + a_3h(x) = 0$ for all x only when $a_1 = a_2 = a_3 = 0$.

(a) Independent.

Step 1: Set up the equation $4a_1 + a_2x^2 + a_3e^{2x} = 0$ for all x .

Step 2: At $x = 0$: $4a_1 + a_2(0) + a_3(1) = 0 \Rightarrow 4a_1 + a_3 = 0$.

Step 3: Take derivative: $0 + 2a_2x + 2a_3e^{2x} = 0$.

Step 4: At $x = 0$ in derivative: $0 + 0 + 2a_3 = 0 \Rightarrow a_3 = 0$.

Step 5: From Step 2 with $a_3 = 0$: $4a_1 = 0 \Rightarrow a_1 = 0$.

Step 6: From Step 3 with $a_3 = 0$: $2a_2x = 0$ for all $x \Rightarrow a_2 = 0$.

Therefore: $a_1 = a_2 = a_3 = 0$ is the only solution. **Independent.**

(b) Independent.

Step 1: Set up $a_1x + a_2x^2 + a_3x^3 = 0$ for all x .

Step 2: Factor out x : $x(a_1 + a_2x + a_3x^2) = 0$ for all x .

Step 3: For this to hold for all x , we need $a_1 + a_2x + a_3x^2 = 0$ for all x .

Step 4: A polynomial equals zero for all x only if all coefficients are zero.

Therefore: $a_1 = a_2 = a_3 = 0$. **Independent.**

(c) Dependent.

Step 1: Observe that $g(x) = 5x = 5 \cdot f(x)$.

Step 2: This gives us the relation: $g(x) - 5f(x) = 0$.

Step 3: In terms of coefficients: $(-5) \cdot f(x) + (1) \cdot g(x) + (0) \cdot h(x) = 0$.

Since we have non-trivial coefficients $(-5, 1, 0)$, the functions are **dependent**.

2 LINEAR INDEPENDENCE OF VECTORS

2.1 Problem Set 2.1: 3D Vector Spaces

Problem 2.1.1

Are the following sets of vectors linearly independent or dependent?

(a) $\vec{A} = (3, 0, 0)$, $\vec{B} = (0, 2, 0)$, $\vec{C} = (0, 0, 1)$

(b) $\vec{A} = (6, 9, 0)$, $\vec{B} = (2, 3, 0)$

(c) $\vec{A} = (2, 3, 1)$, $\vec{B} = (0, 1, 2)$, $\vec{C} = (0, 0, 5)$

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Solution 2.1.1

Method: Vectors are linearly independent if $a_1\vec{A} + a_2\vec{B} + \dots = \vec{0}$ only when all coefficients are zero.

(a) Independent.

Step 1: Set up equation: $a\vec{A} + b\vec{B} + c\vec{C} = \vec{0}$.

Step 2: Substitute: $a(3, 0, 0) + b(0, 2, 0) + c(0, 0, 1) = (0, 0, 0)$.

Step 3: Expand: $(3a, 2b, c) = (0, 0, 0)$.

Step 4: Component equations:

- $3a = 0 \Rightarrow a = 0$
- $2b = 0 \Rightarrow b = 0$
- $c = 0 \Rightarrow c = 0$

Step 5: Only trivial solution exists.

Therefore: **Independent**. These form a coordinate basis.

(b) Dependent.

Step 1: Check proportionality: $\frac{6}{2} = 3$ and $\frac{9}{3} = 3$.

Step 2: Both vectors have zero in third component.

Step 3: Therefore: $\vec{A} = 3\vec{B}$.

Step 4: This gives: $\vec{A} - 3\vec{B} = \vec{0}$.

Step 5: Non-trivial coefficients $(1, -3)$ exist.

Therefore: **Dependent**.

(c) Independent.

Step 1: Set up: $a\vec{A} + b\vec{B} + c\vec{C} = \vec{0}$.

Step 2: Substitute: $a(2, 3, 1) + b(0, 1, 2) + c(0, 0, 5) = (0, 0, 0)$.

Step 3: Expand: $(2a, 3a + b, a + 2b + 5c) = (0, 0, 0)$.

Step 4: System of equations:

- $2a = 0 \Rightarrow a = 0$
- $3a + b = 0 \Rightarrow b = 0$ (since $a = 0$)
- $a + 2b + 5c = 0 \Rightarrow c = 0$ (since $a = b = 0$)

Step 5: Only trivial solution exists.

Therefore: **Independent**.

3 DIRAC NOTATION (BRA-KET FORMALISM)

3.1 Physical Interpretation of Scalar Products

“In physics, you don’t have to go around making trouble for yourself. Nature does it for you.” — Frank Wilczek

Physical Meaning of the Scalar Product

The scalar product $\langle\phi|\psi\rangle$ has two fundamental interpretations:

- 1. Geometric Interpretation:** The product $\langle\phi|\psi\rangle$ represents the projection of $|\psi\rangle$ onto $|\phi\rangle$.
- 2. Probabilistic Interpretation:** For normalized states, $\langle\phi|\psi\rangle$ represents the probability amplitude that a system in state $|\psi\rangle$ will be found in state $|\phi\rangle$ upon measurement.

Key Points:

- The probability amplitude is generally complex
- The actual probability is $|\langle\phi|\psi\rangle|^2$
- For normalized states: $\langle\psi|\psi\rangle = 1$

3.2 Problem Set 3.1: Basic Bra-Ket Operations

Problem 3.1.1

Consider the kets:

$$|\psi\rangle = \begin{pmatrix} 3i \\ 2-i \\ 4 \end{pmatrix}, \quad |\phi\rangle = \begin{pmatrix} 2 \\ i \\ 2+3i \end{pmatrix}$$

- (a) Find $\langle\phi|$ and calculate $\langle\phi|\psi\rangle$
- (b) Find the probability of measuring state $|\psi\rangle$ in state $|\phi\rangle$
- (c) Interpret the result physically

Solution 3.1.1**(a) Calculate inner product**

Step 1: Find $\langle\phi|$ by taking complex conjugate transpose:

$$\langle\phi| = (2^* \quad i^* \quad (2+3i)^*) = (2 \quad -i \quad 2-3i)$$

Step 2: Calculate $\langle\phi|\psi\rangle$:

$$\langle\phi|\psi\rangle = (2 \quad -i \quad 2-3i) \begin{pmatrix} 3i \\ 2-i \\ 4 \end{pmatrix}$$

Step 3: Expand each term:

- $2 \cdot (3i) = 6i$
- $(-i) \cdot (2-i) = -2i + i^2 = -2i - 1$
- $(2-3i) \cdot 4 = 8 - 12i$

Step 4: Sum: $\langle\phi|\psi\rangle = 6i + (-2i - 1) + (8 - 12i) = 7 - 8i$

(b) Calculate transition probability

Step 5: Calculate norms for normalization:

- $\langle\psi|\psi\rangle = |3i|^2 + |2-i|^2 + |4|^2 = 9 + 5 + 16 = 30$
- $\langle\phi|\phi\rangle = |2|^2 + |-i|^2 + |2+3i|^2 = 4 + 1 + 13 = 18$

Step 6: Transition probability between unnormalized states:

$$P = \frac{|\langle\phi|\psi\rangle|^2}{\langle\phi|\phi\rangle \cdot \langle\psi|\psi\rangle} = \frac{|7-8i|^2}{18 \cdot 30} = \frac{49+64}{540} = \frac{113}{540}$$

(c) Physical interpretation

The complex amplitude $7 - 8i$ contains both magnitude and phase information. The probability $\frac{113}{540} \approx 0.209$ means there's about a 21% chance of finding the system in state $|\phi\rangle$ if measured after preparation in state $|\psi\rangle$.

4 HERMITIAN CONJUGATE OPERATIONS

4.1 Problem Set 4.1: Basic Hermitian Conjugates

Problem 4.1.1

Find the Hermitian conjugate of the following:

(a) $|\psi\rangle = \begin{pmatrix} 1 + 2i \\ 3 - i \\ 2i \end{pmatrix}$

(b) $\alpha = 3 + 4i$

(c) The matrix $A = \begin{pmatrix} 1 & 2 + i \\ 3 - 2i & 4 \end{pmatrix}$

Solution 4.1.1

Rule: The Hermitian conjugate involves taking the complex conjugate and transpose: $A^\dagger = (A^*)^T$.

(a) Vector (ket to bra)

Step 1: Take complex conjugate of each component:

- $(1 + 2i)^* = 1 - 2i$
- $(3 - i)^* = 3 + i$
- $(2i)^* = -2i$

Step 2: Transpose (column to row):

$$|\psi\rangle^\dagger = \langle\psi| = (1 - 2i \quad 3 + i \quad -2i)$$

(b) Scalar (complex number)

Step 1: For scalars, Hermitian conjugate is just complex conjugate.

Step 2: $\alpha^\dagger = (3 + 4i)^* = 3 - 4i$

(c) Matrix

Step 1: Take complex conjugate of each element:

$$A^* = \begin{pmatrix} 1 & 2 - i \\ 3 + 2i & 4 \end{pmatrix}$$

Step 2: Take transpose:

$$A^\dagger = (A^*)^T = \begin{pmatrix} 1 & 3 + 2i \\ 2 - i & 4 \end{pmatrix}$$

5 HERMITIAN OPERATORS

5.1 Theory: Properties of Hermitian Operators

Definition: Hermitian Operator

An operator $\hat{A}$ is called Hermitian (or self-adjoint) if it satisfies:

$$\hat{A}^\dagger = \hat{A}$$

Equivalently, for any states $|\psi\rangle$ and $|\phi\rangle$:

$$\langle\psi|\hat{A}|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*$$

Fundamental Properties of Hermitian Operators

1. **Real Eigenvalues:** All eigenvalues of a Hermitian operator are real numbers.
2. **Orthogonal Eigenvectors:** Eigenvectors corresponding to different eigenvalues are orthogonal.
3. **Complete Orthonormal Basis:** The eigenvectors form a complete orthonormal basis.
4. **Real Expectation Values:** For any state $|\psi\rangle$: $\langle\psi|\hat{A}|\psi\rangle \in \mathbb{R}$

Physical Significance: Hermitian operators represent observable quantities in quantum mechanics.

5.2 Problem Set 5.1: Hermitian Operator Problems

Problem 5.1.1

Verify that the Pauli matrices are Hermitian and find their eigenvalues:

(a) $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(b) $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

(c) $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Solution 5.1.1**(a) Pauli-X matrix**

Step 1: Check if Hermitian by verifying $\sigma_x^\dagger = \sigma_x$:

$$\sigma_x^\dagger = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x$$

Step 2: Find eigenvalues using characteristic equation $\det(\sigma_x - \lambda I) = 0$:

$$\det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - 1 = 0$$

Step 3: Eigenvalues: $\lambda = \pm 1$

(b) Pauli-Y matrix

Step 1: Check if Hermitian:

$$\sigma_y^\dagger = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}^\dagger = \begin{pmatrix} 0 & (i)^* \\ (-i)^* & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma_y$$

Step 2: Find eigenvalues:

$$\det \begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix} = (-\lambda)(-\lambda) - (-i)(i) = \lambda^2 - (-i^2) = \lambda^2 - 1 = 0$$

Step 3: Eigenvalues: $\lambda = \pm 1$

(c) Pauli-Z matrix

Step 1: Check if Hermitian:

$$\sigma_z^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z$$

Step 2: Find eigenvalues:

$$\det \begin{pmatrix} 1 - \lambda & 0 \\ 0 & -1 - \lambda \end{pmatrix} = (1 - \lambda)(-1 - \lambda) = -(1 - \lambda^2) = \lambda^2 - 1 = 0$$

Step 3: Eigenvalues: $\lambda = \pm 1$

Conclusion: All Pauli matrices are Hermitian with real eigenvalues $\{+1, -1\}$.

6 UNITARY MATRICES

6.1 Theory: Properties of Unitary Matrices

Definition: Unitary Matrix

A matrix U is called unitary if it satisfies:

$$U^\dagger U = UU^\dagger = I$$

Equivalent conditions:

- $U^\dagger = U^{-1}$ (Hermitian conjugate equals inverse)
- The columns of U form an orthonormal set
- The rows of U form an orthonormal set

Fundamental Properties of Unitary Matrices

1. **Norm Preservation:** For any vector $|v\rangle$: $\|U|v\rangle\| = \||v\rangle\|$
2. **Inner Product Preservation:** $\langle Uu|Uv\rangle = \langle u|v\rangle$
3. **Eigenvalues on Unit Circle:** All eigenvalues λ satisfy $|\lambda| = 1$
4. **Determinant:** $|\det(U)| = 1$

Physical Significance: Unitary matrices represent reversible quantum operations that preserve probability amplitudes.

6.2 Problem Set 6.1: Unitary Matrix Problems

Problem 6.1.1 (Basic): Simple Unitary Matrices

Verify that the following matrices are unitary:

(a) Identity: $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(b) Bit-flip: $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(c) Phase gate: $P = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$

Solution 6.1.1

Method: Verify $U^\dagger U = I$ for each matrix.

(a) Identity matrix

Step 1: $I^\dagger = I$ (real matrix, so conjugate transpose = transpose = original)

Step 2: $I^\dagger I = I \cdot I = I$

(b) Bit-flip matrix

Step 1: $X^\dagger = X$ (real matrix)

Step 2: $X^\dagger X = X^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

(c) Phase gate

Step 1: $P^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & i^* \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$

Step 2: $P^\dagger P = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & (-i)(i) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

All three matrices are confirmed unitary.

Problem 6.1.2 (Intermediate): Hadamard Gate

The Hadamard gate is defined as:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

(a) Show that H is unitary

(b) Calculate $H|0\rangle$ and $H|1\rangle$ where $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

(c) Show that $H^2 = I$

Solution 6.1.2**(a) Prove unitarity**

Step 1: Since H is real, $H^\dagger = H^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H$

Step 2: Calculate $H^\dagger H = H^2$:

$$H^2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Step 3: Matrix multiplication:

$$H^2 = \frac{1}{2} \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

(b) Action on basis states

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

(c) Self-inverse property

This was already proven in part (a): $H^2 = I$

Physical interpretation: The Hadamard gate creates equal superposition states and is its own inverse.

Note: Why $\frac{1}{\sqrt{2}}$?

The factor $\frac{1}{\sqrt{2}}$ ensures normalization. For the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$:

$$\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} + \frac{1}{2} = 1$$

This preserves the unit length of quantum states, which is required for probability conservation.

7 UNITARY OPERATORS AND TRANSFORMATIONS

7.1 Problem Set 7.1: Sequential Unitary Operations

Problem 7.1.1 (Basic): Composition of Gates

Consider applying the following sequence of gates to the initial state $|0\rangle$:

1. Apply Hadamard gate H
 2. Apply Pauli-X gate X
 3. Apply Hadamard gate H again
- (a) Calculate the state after each step
(b) What is the final state?
(c) What is the overall transformation matrix for this sequence?

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Solution 7.1.1**Initial state:** $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ **(a) Step-by-step calculation****After Step 1 (Apply H):**

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

After Step 2 (Apply X):

$$X \left[\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Note: The state remains unchanged because X swaps components, but both components are equal.

After Step 3 (Apply H again):

$$H \left[\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

(b) Final state

The final state is $|0\rangle$ - we return to the initial state.

(c) Overall transformation matrix

The overall transformation is: HXH

$$HXH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

First calculate XH :

$$XH = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

Then calculate $H(XH)$:

$$HXH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$$

The sequence $HXH = Z$ (Pauli-Z gate).

Note: Quantum Gate Identity

The identity $HXH = Z$ is an important quantum gate relation. It shows that:

- We can implement a Z-gate using H and X gates
- Different gate sequences can produce the same overall effect
- This is useful in quantum computing when certain gates are easier to implement physically

Problem 7.1.2 (Intermediate): General Rotation Gates

A rotation gate around the y-axis by angle θ is given by:

$$R_y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

- (a) Show that $R_y(\theta)$ is unitary
- (b) Find $R_y(\pi)$ and compare with known Pauli gates
- (c) What state does $R_y(\pi/2)|0\rangle$ produce?
- (d) Show that $R_y(\theta_1)R_y(\theta_2) = R_y(\theta_1 + \theta_2)$

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Solution 7.1.2**(a) Prove unitarity**

$$R_y(\theta)^\dagger = \begin{pmatrix} \cos(\theta/2) & \sin(\theta/2) \\ -\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

Calculate $R_y(\theta)^\dagger R_y(\theta)$:

$$\begin{aligned} & \begin{pmatrix} \cos(\theta/2) & \sin(\theta/2) \\ -\sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \\ &= \begin{pmatrix} \cos^2(\theta/2) + \sin^2(\theta/2) & 0 \\ 0 & \sin^2(\theta/2) + \cos^2(\theta/2) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{aligned}$$

(b) Special case: $R_y(\pi)$

$$R_y(\pi) = \begin{pmatrix} \cos(\pi/2) & -\sin(\pi/2) \\ \sin(\pi/2) & \cos(\pi/2) \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = iY$$

This is i times the Pauli-Y matrix (up to a global phase).

(c) Action on $|0\rangle$

$$\begin{aligned} R_y(\pi/2)|0\rangle &= \begin{pmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \end{aligned}$$

This produces an equal superposition state.

(d) Composition property

Using angle addition formulas:

$$\cos((\theta_1 + \theta_2)/2) = \cos(\theta_1/2) \cos(\theta_2/2) - \sin(\theta_1/2) \sin(\theta_2/2) \quad (1)$$

$$\sin((\theta_1 + \theta_2)/2) = \sin(\theta_1/2) \cos(\theta_2/2) + \cos(\theta_1/2) \sin(\theta_2/2) \quad (2)$$

Matrix multiplication $R_y(\theta_1)R_y(\theta_2)$ yields:

$$\begin{aligned} & \begin{pmatrix} \cos(\theta_1/2) \cos(\theta_2/2) - \sin(\theta_1/2) \sin(\theta_2/2) & -\cos(\theta_1/2) \sin(\theta_2/2) - \sin(\theta_1/2) \cos(\theta_2/2) \\ \sin(\theta_1/2) \cos(\theta_2/2) + \cos(\theta_1/2) \sin(\theta_2/2) & \cos(\theta_1/2) \cos(\theta_2/2) - \sin(\theta_1/2) \sin(\theta_2/2) \end{pmatrix} \\ &= \begin{pmatrix} \cos((\theta_1 + \theta_2)/2) & -\sin((\theta_1 + \theta_2)/2) \\ \sin((\theta_1 + \theta_2)/2) & \cos((\theta_1 + \theta_2)/2) \end{pmatrix} = R_y(\theta_1 + \theta_2) \end{aligned}$$

This confirms that rotations compose additively.

“The most beautiful thing we can experience is the mysterious. It is the source of all true art and science.” — Albert Einstein

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For Quantum Entanglement and Quantum Computing
Mathematical foundations for the quantum future

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